UNIVERSITY OF CAMBRIDGE

# NATURAL SCIENCES TRIPOS Part II

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Tuesday 2nd June 2026 09:00 to 11:00

PHYSICS
PHYSICAL SCIENCES: HALF SUBJECT PHYSICS

ADVANCED AND QUANTUM OPTICS

*Candidates offering this paper should attempt* ***all six*** *questions (four from Section A and two from Section B).*

*The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.*

| STATIONERY REQUIREMENTS | SPECIAL REQUIREMENTS |
|---|---|
| $2 \times$ 8-page answer booklet | Mathematical Formulae Handbook |
| Rough workpad | Approved calculator allowed |
| Yellow master coversheet | |

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

## SECTION A

*Attempt* ***all four*** *questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.*

**1** A Gaussian beam from a He–Ne laser ($\lambda = 633$ nm) has a beam waist $W_0 = 0.5$ mm. Calculate the Rayleigh range $z_0$, the beam divergence half-angle $\theta_0$, and the beam diameter at a distance of 10 m from the waist.

**2** In a medium with a second-order nonlinear susceptibility $\chi^{(2)}$, two monochromatic plane waves at frequencies $\omega_1$ and $\omega_2$ co-propagate with wavevectors $\mathbf{k}_1$ and $\mathbf{k}_2$. State the frequency-matching and phase-matching conditions for sum-frequency generation, and explain physically why the phase-matching condition is necessary for efficient conversion. Briefly describe one practical method by which phase matching can be achieved in a uniaxial birefringent crystal.

**3** A planar dielectric waveguide has a core of refractive index $n_1 = 1.50$ and thickness $d = 5.0$ $\mu$m, surrounded by cladding of refractive index $n_2 = 1.48$. Calculate the numerical aperture and the number of TE modes supported at a free-space wavelength $\lambda_0 = 1.55$ $\mu$m.

**4** An optical resonator of length $d = 20$ cm consists of two mirrors with intensity reflectances $\mathcal{R}_1 = 0.99$ and $\mathcal{R}_2 = 0.97$. Assume the refractive index $n = 1$ and negligible internal losses ($\alpha_s = 0$). Calculate the free spectral range $\nu_F$, the finesse $\mathcal{F}$, and the photon lifetime $\tau_p$.

## SECTION B

*Attempt **both** questions from this Section.*

**5** This question concerns lasers and their spectral properties.

(a) Consider an atom with two energy levels separated by $h\nu_0$, placed in a cavity of volume $V$ containing $n$ photons in a mode of frequency $\nu \approx \nu_0$. Write down expressions for the probability densities of absorption, stimulated emission, and spontaneous emission in terms of $n$, $c$, $V$, and the transition cross section $\sigma(\nu)$.

(b) A laser amplifier is pumped so that atoms are excited into level 2 at rate $R_2$ (per unit volume per unit time), and depumped from level 1 at rate $R_1$. In the absence of amplifier radiation, the rate equations for the population densities are

$$\frac{dN_2}{dt} = R_2 - \frac{N_2}{\tau_2}\,, \qquad \frac{dN_1}{dt} = -R_1 - \frac{N_1}{\tau_1} + \frac{N_2}{\tau_{21}}\,.$$

Solve these for the steady-state population difference $N_0 = N_2 - N_1$ and discuss the conditions on the lifetimes $\tau_1$, $\tau_2$, and $\tau_{21}$ that favour a large population inversion.

(c) Now include the effect of amplifier radiation, characterised by the stimulated transition rate $W_i = \phi\,\sigma(\nu)$. Show that the steady-state population difference becomes

$$N = \frac{N_0}{1 + \tau_s W_i}\,,$$

identifying the saturation time constant $\tau_s$ and the saturation photon-flux density $\phi_s(\nu)$.

(d) A laser oscillator uses the above gain medium inside a Fabry–Perot resonator of length $d$ with loss coefficient $\alpha_r$. State the gain and phase conditions for laser oscillation. By equating the saturated gain to the loss, derive an expression for the steady-state intracavity photon-flux density above threshold in terms of $\phi_s$, $\gamma_0(\nu)$, and $\alpha_r$.

(e) Explain, with the aid of sketches, how the number of oscillating modes differs between a homogeneously broadened and an inhomogeneously broadened gain medium. Define the term *spectral hole burning.*

**6** This question considers coherence, interferometry, and nonlinear optics.

(a) Define the complex degree of temporal coherence $g(\tau)$ and the coherence time $\tau_c$. For a light source with a Lorentzian spectral profile of FWHM $\Delta\nu$, state (without proof) the form of $|g(\tau)|$ and give the resulting coherence length $l_c$.

(b) A Mach–Zehnder interferometer is illuminated with quasi-monochromatic light of mean frequency $\bar{\nu}$. One arm introduces a variable delay $\tau$. Show that the output intensity can be written as

$$I(\tau) = I_0\big[1 + |g(\tau)|\cos(2\pi\bar{\nu}\tau)\big]\,,$$

and hence explain how the fringe visibility provides a direct measurement of $|g(\tau)|$.

(c) A monochromatic Gaussian beam of wavelength $\lambda$ and peak intensity $I_0$ passes through a thin slab of glass (thickness $d_{\mathrm{g}}$, linear refractive index $n$) that has a finite third-order nonlinear coefficient $n_2$. The beam intensity profile is $I(r) = I_0\exp(-2r^2/w^2)$. Show that the nonlinear phase acquired across the beam profile acts as a thin lens, and derive an expression for the focal length of this effective lens in terms of $n_2$, $I_0$, $d_{\mathrm{g}}$, $w$, and $\lambda$.

(d) A lithium niobate crystal is used as a Pockels cell (electro-optic modulator). Explain how applying a DC electric field $E_0$ to a medium with $\chi^{(2)}$ modifies the refractive index, and derive the general expression $n(E_0) \approx n_0 + \frac{\chi^{(2)}}{2n_0}E_0$. If the crystal of length $L$ is placed between crossed polarisers at 45° to its principal axes, show that the transmitted intensity varies as $I_{\mathrm{out}} = I_{\mathrm{in}}\sin^2\!\big(\pi V/2V_\pi\big)$, identifying the half-wave voltage $V_\pi$.

END OF PAPER